

Zhengxu Yu

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RESEARCH OVERVIEW

My research concerns the generalisation limits of learning systems: how far competence acquired from finite experience extends beyond the distribution that produced it. Existing accounts of distribution shift buy their guarantees with assumptions that cannot be checked at deployment, and the methods built on them have proved hard to separate from plain empirical risk minimisation. My interest is in the case they do not reach, where the new regime is not a perturbation of the training one and no amount of interpolation within what was seen will produce the right behaviour. What is missing there is exploration that can leave the region the training data describes rather than search inside it, and I take that to depend on how the agent represents its environment. Dynamics learned as executable programs, written from the agent's own interaction and revised wherever prediction and observation disagree, form a hypothesis that can be rejected outright and replaced in a way a distributed representation does not allow. Whether this is what governs behaviour in an unfamiliar regime is a question about held-out environments and has to be settled by running there: currently on ARC-AGI-3, and in self-organising multi-agent populations.

EDUCATION

Ph.D. Computer Science 2017–2021	Zhejiang University Advisors: Prof. Deng Cai and Prof. Xiaofei He Thesis: Stable Learning for Non-I.I.D Data
M.Sc. Computer Science 2015–2016	University of Surrey Advisor: Prof. H. Lilian Tang Thesis: CNN-based Mycobacterium Cells Segmentation for Time-lapse Images
B.Eng. Electrical Engineering 2011–2015	Jilin University

EMPLOYMENT

Research Scientist Jan 2026–Present	Huawei Technologies R&D UK, London - Built a generative code world model in which agents explore open-ended environments and encode the rules they discover as executable, self-improving code, reaching 100% Relative Human Action Efficiency (RHAE) on the ARC-AGI-3 community leaderboard. - Led research and system design for OneManCompany (OMC) , an open-source agentic AI platform for self-organising multi-agent systems, leading a 5-person team; 400+ GitHub stars, with the design published as arXiv:2604.22446. https://github.com/1mancompany/OneManCompany
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Member of Technical Team Aug 2025–Jan 2026	Inephany Ltd., London - Built the multi-agent reinforcement learning training infrastructure for large language models and vision transformers, supporting distributed training runs on clusters of 500+ GPUs.
Algorithm Expert 2024–2025	Apsara Lab (now Tongyi Lab), Alibaba Cloud - Developed RL-based post-training methods for LLM reasoning efficiency, alignment and agentic tool use, cutting reasoning-token usage by 50% on Qwen-7B at no accuracy loss (NeurIPS 2025, AACL 2026). - Delivered three bespoke LLM agent systems with dynamic reasoning and tool use for decision support and information retrieval, deployed for the International Olympic Committee and several municipal government bureaus in China. - Shipped those three systems and two further large-scale enterprise solutions as tech lead and core algorithm owner, contributing to over RMB 25M in cumulative GAAP-recognised revenue. - Led a 13-engineer feature team and a 40-person data-annotation group, supervising 8 research interns and junior engineers.
Senior Algorithm Engineer 2021–2024	Apsara Lab (now Tongyi Lab), Alibaba Cloud - Developed multi-agent reinforcement learning for city-scale traffic signal control, reducing mean travel time by 2.8% against the CoLight state of the art on real-city road networks of up to 196 intersections, and deployed it in production urban traffic signal control (IJCAI, IEEE TKDE, IJMLC). - Developed progressive transfer learning under distribution shift, raising Rank-1 accuracy by up to 4.5 points over the MGN backbone across four person re-identification benchmarks (MSMT17, Market-1501, CUHK03, DukeMTMC-reID) (IEEE TIP).
Research Intern 2018–2021	DAMO Academy, Alibaba Group - Developed large-scale multi-agent reinforcement learning and deep representation learning methods for the traffic engine of Alibaba City Brain; 7 granted national invention patents.

SELECTED PUBLICATIONS

The * symbol denotes equal contribution.

Conference Papers

- C4** Weihang Pan, Zhengxu Yu, Yuxiang Zhang, Zhongming Jin, Binbin Lin, Xiaofei He, Jieping Ye. “ChainPrune: Evaluating and Reducing Redundancy in Long Chain-of-Thought Reasoning” *ISWC*, 2026.
- C3** Weihang Pan, Zhengxu Yu, Yong Wu, Xun Liang, Zhongming Jin, Qiang Fu, Penghui Shang, Binbin Lin, Xiaofei He, Jieping Ye. “FGD-Align: Pluralistic Alignment for Large Language Models via Fuzzy Group Decision-Making” *AAAI Conference on Artificial Intelligence (AAAI)*, 2026.
- C2** Yuxiang Zhang*, Zhengxu Yu*, Weihang Pan, Zhongming Jin, Qiang Fu, Deng Cai, Binbin Lin, Jieping Ye. “TokenSqueeze: Performance-Preserving Compression for Reasoning LLMs” *Neural Information Processing Systems (NeurIPS)*, 2025.

- C1** Zhengxu Yu*, Shuxian Liang*, Long Wei, Zhongming Jin, Jianqiang Huang, Deng Cai, Xiaofei He, Xian-Sheng Hua. “MaCAR: Urban Traffic Light Control via Active Multi-agent Communication and Action Rectification” *International Joint Conference on Artificial Intelligence (IJCAI)*, 2020.

Journal Articles

- J4** Weihang Pan, Binbin Lin, Yafei Wang, Zhengxu Yu, Xinkui Zhao, Xiaofei He, Jieping Ye. “Cooperative Driving at Multiple Unsignalized Intersections in Fully Autonomous Driving Scenarios” *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, vol. 26, no. 12, 2025.
- J3** Chao Xiang, Zhongming Jin, Zhengxu Yu, Xian-Sheng Hua, Yao Hu, Wei Qian, Kaili Zhu, Deng Cai, Xiaofei He. “Optimizing Traffic Efficiency via a Reinforcement Learning Approach Based on Time Allocation” *International Journal of Machine Learning and Cybernetics (IJMLC)*, vol. 14, no. 10, 2023.
- J2** Xin Guo, Zhengxu Yu, Pengfei Wang, Zhongming Jin, Jianqiang Huang, Deng Cai, Xiaofei He, Xian-Sheng Hua. “Urban Traffic Light Control via Active Multi-Agent Communication and Supply-Demand Modeling” *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, vol. 35, no. 4, 2023. (Journal extension of C1.)
- J1** Zhengxu Yu, Dong Shen, Zhongming Jin, Jianqiang Huang, Deng Cai, Xian-Sheng Hua. “Progressive Transfer Learning” *IEEE Transactions on Image Processing (TIP)*, vol. 31, 2022.

Technical Reports

- P1** Zhengxu Yu, Yu Fu, Zhiyuan He, Yuxuan Huang, Ka Yiu Lee, Meng Fang, Weilin Luo, Jun Wang. “OneManCompany: An Open-Source Operating System for Self-Organizing Multi-Agent Systems” *arXiv preprint arXiv:2604.22446*, 2026. [Paper](#).